Timothy Do

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Education

University of California, Los Angeles — Sep. 2023 – Dec. 2024 *(Expected)*
M.S. Electrical and Computer Engineering, Signal Processing Specialization — *Los Angeles, CA*

- **Cumulative GPA:** 3.90/4.0, **Specialization GPA:** 4.0/4.0
- **Relevant Coursework:** Computational Imaging, Digital Speech Processing, Deep Learning & Neural Networks
- **Clubs and Societies:** ECEGAPS (Electrical and Computer Engineering Graduate and PostDoc Society)

University of California, Irvine — Sep. 2019 – June 2023
B.S. Electrical Engineering, Accelerated Status — *Irvine, CA*

- **Cumulative GPA:** 3.924/4.0, **Major GPA:** 3.944/4.0, **Magna Cum Laude**
- **Specializations:** Digital Signal Processing, Communications
- **Relevant Coursework:** Computer Architecture, Control Systems, Digital Image/Signal Processing, Electronics
- **Clubs and Societies:** Irvine Computer Vision Laboratory, IEEE, IPC, Eta Kappa Nu, Tau Beta Pi

Experience

Embedded Software Engineering Intern — June 2023 – Present
Maxeon Solar Technologies — *San Jose, CA*

- Implemented Robust OpenCV pipelines to quantitatively detect module features for defect binning
- Developed Tkinter GUI interfaces for operator ease of use when testing images under custom data pipelines
- Performed Multi-Class Data Labeling and Implemented Data Augmentation methods for training robust YOLOv5/YOLOv8 Defect Classification Models

Lead Undergraduate Student Researcher — January 2022 – June 2023
Irvine Computer Vision Laboratory at UC Irvine — *Irvine, CA*

- Researched under Professor Glenn Healey on the topic of Spectral Image Filtering and Baseball Simulations
- Digitized Hyperspectral Optical Imaging Filters into CSVs and compared them against certain chemical emissions
- Managed 15+ Graduate/Undergraduate Researchers in Google Spaces and developed shared MATLAB/Python Scripts hosted on Google Drive/Github for analyzing SWIR/NVIR vegetation spectras

Industrial Internet of Things Intern — June 2022 – Sep. 2022
Western Digital — *San Jose, CA*

- Applied Industrial Toolsets to develop a Raspberry Pi Sensor Data Analytics Platform
- Earned the Six Sigma Yellow Belt for learning foundations of Lean Six Sigma
- Saved the company an annual projected cost of $312,000 from hazard mitigations

Projects

RuHuman Audio Verification System | *Python, MATLAB, Javascript, Librosa, Flask* — Sep. 2023 – Dec. 2023

- Developed a resilient audio verification system to classify between bonafide and spoofed speech
- Investigated the effects of AWGN on the ASVSpoof2021 set based on t-DCF & EER metrics
- Implemented a Flask Server s record/upload audio samples for verification from any web device with a microphone

The Do-Pro Senior Design Project | *Raspberry Pi, Python, MATLAB, OpenCV* — Sep. 2022 - March 2023

- Led a team of four in creating an embedded stereo-vision camera for ADAS detection and 3D-Reconstruction
- Calibrated the stereo-camera in MATLAB and implemented block matching algorithms with OpenCV
- Designed the button/touch-screen user interface using RPi.GPIO and Tkinter for navigating the camera
- Developed a Power Regression Model for the Do-Pro to percieve depth with 6% error from 1-3 feet

EECS195-Colorization | *Python, MATLAB, OpenCV, Tensorflow* — Feb. 2022 – March 2022

- Colorized grayscale images using GAN and Auto-Encoder Neural Networks with finetuning on custom data
- Gathered custom datasets by taking videos of different scenery with distinct objects then split them into frames
- Clustered pixels into superpixel subgraphs for uniform training weights during preprocessing

Skills

Electronics: Raspberry Pi, Network Analysis, Cadence, LTSpice, Soldering, Oscilloscope & Multimeter Measurements
Programming Languages: Python, Java, C, CUDA, MATLAB, bash, MySQL HTML/CSS/Javascript, VHDL
Developer Tools: git, SSH, Makefile, Linux, Azure, VS Code, Visual Studio, PyCharm, Eclipse, Vivado HLS
Libraries: OpenCV, Pandas, Numpy, Matplotlib, PIL, Tensorflow, Pytorch, Tkinter, Socket, Flask, Librosa